

ISyE6669-OAN Homework Week 10

Fall 2025

1. In this problem, we want to walk you through the column generation algorithm to solve the cutting stock problem. Consider the following formulation

$$\begin{aligned} \min \quad & \sum_{j=1}^n x_j \\ \text{s.t.} \quad & \sum_{j=1}^N \mathbf{A}_j x_j = \mathbf{b} \\ & x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

The problem has the following data. Customers need three types of smaller widths: $w_1 = 20, w_2 = 35, w_3 = 45$ with quantities $b_1 = 25, b_2 = 15, b_3 = 10$. The width of a big roll is $W = 100$.

- (a) Assume the column generation algorithm starts from the following initial patterns:

$$\mathbf{A}_1 = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}, \mathbf{A}_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \mathbf{A}_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}.$$

Write down the restricted master problem (RMP) using these patterns. Solve this RMP by hand. Find the optimal basis $\mathbf{B}$ and its inverse $\mathbf{B}^{-1}$. Find the optimal dual solution $\hat{\mathbf{y}}^\top = \mathbf{c}_B^\top \mathbf{B}^{-1}$.

Solutions: Using given initial patterns we get following RMP:

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 \\ \text{s.t.} \quad & 5x_1 = 25 \\ & 2x_2 = 15 \\ & 2x_3 = 10 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Optimal solution of this RMP is:

$$x = \begin{bmatrix} 25/5 \\ 15/2 \\ 10/2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7.5 \\ 5 \end{bmatrix}$$

and its optimal value is $5 + 7.5 + 5 = 17.5$

Optimal basis and its inverse are:

$$B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1/5 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$$

From there we obtain optimal dual solution:

$$\begin{aligned} \hat{y}^\top &= c_B^\top B^{-1} = [1 \quad 1 \quad 1] \begin{bmatrix} 1/5 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{bmatrix} = \\ &= [1/5 \quad 1/2 \quad 1/2] = \\ &= [0.2 \quad 0.5 \quad 0.5] \end{aligned}$$

- (b) Solve RMP in Python CVX. Write down the optimal solution, the optimal basis B , and its inverse B^{-1} . Find the optimal dual solution $\hat{y}^\top = c_B^\top B^{-1}$. To take the inverse of B , you can use a calculator or computer program. In this iteration, you should be able to solve this LP by hand. But we ask you to set up the code in CVX and solve it using CVX. This code will be used in later iterations.

Solution: By solving above RMP in CVXPY we obtain following optimal solution:

$$x = \begin{bmatrix} 5 \\ 7.5 \\ 5 \end{bmatrix}$$

and optimal value of 17.5.

Since all three variables are nonzero, they are all basic and hence basic matrix and its inverse are (as expected same as in part 1.):

$$B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 0.2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

As before, we obtain optimal dual solution:

$$\begin{aligned} \hat{y}^\top &= c_B^\top B^{-1} = [1 \quad 1 \quad 1] \begin{bmatrix} 0.2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix} = \\ &= [0.2 \quad 0.5 \quad 0.5] \end{aligned}$$

- (c) Write down the pricing problem, i.e. the knapsack problem using the above data and the optimal dual solution you found.

Solution: Using $\hat{y}$ we obtain following knapsack/pricing problem:

$$\begin{aligned} \max \quad & \frac{1}{5}a_1 + \frac{1}{2}a_2 + \frac{1}{2}a_3 \\ \text{s.t.} \quad & 20a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integer} \end{aligned}$$

- (d) Solve the pricing problem in CVX. Should we terminate the column generation algorithm at this point? Explain. If the column generation should continue, what is the new pattern generated by the pricing problem?

Solution: Using CVXPY we obtain following optimal solution of the pricing/knapsack problem:

$$A_4 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

with optimal value $\hat{z} = 1.2$. Since $1 - \hat{z} = -0.2 < 0$, we continue the algorithm.

- (e) If the column generation should continue, then augment (RMP) with the new column and solve it in CVX again. You can easily modify your CVX code by incorporating the new column. Write down the optimal solution, the optimal basis $\mathbf{B}$, and the inverse $\mathbf{B}^{-1}$. Compute the dual variable. Then solve the pricing problem again by modifying the data in your code. Should you terminate the column generation at this iteration? Explain. If the column generation should continue, do the same for all the following iterations until the column generation terminates.

Solution:

Iteration 2: By adding pattern A_4 we get RMP:

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 + x_4 \\ \text{s.t.} \quad & 5x_1 + x_4 = 25 \\ & 2x_2 + 2x_4 = 15 \\ & 2x_3 = 10 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Its optimal value is 16.0 and its optimal solution is:

$$x = \begin{bmatrix} 3.5 \\ 0 \\ 5 \\ 7.5 \end{bmatrix}.$$

We see that basic variables are x_1, x_3 and x_4 . Hence:

$$B = \begin{bmatrix} 5 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 0.2 & -0.1 & 0 \\ 0 & 0 & 0.5 \\ 0 & 0.5 & 0 \end{bmatrix}$$

From there we get:

$$\begin{aligned} \hat{y}^\top &= c_B^\top B^{-1} = [1 \quad 1 \quad 1] \begin{bmatrix} 0.2 & -0.1 & 0 \\ 0 & 0 & 0.5 \\ 0 & 0.5 & 0 \end{bmatrix} = \\ &= [0.2 \quad 0.4 \quad 0.5] = \\ &= [1/5 \quad 2/5 \quad 1/2] \end{aligned}$$

Using this dual optimal solution we get following pricing/knapsack problem:

$$\begin{aligned} \max \quad & 0.2a_1 + 0.4a_2 + 0.5a_3 \\ \text{s.t.} \quad & 20a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integer} \end{aligned}$$

By solving this problem, we get new pattern:

$$A_5 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

with optimal value $\hat{z} = 1.1$. Since $1 - \hat{z} = -0.1 < 0$ we continue the algorithm.

Iteration 3: By adding pattern A_4 we get RMP:

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 + x_4 + x_5 \\ \text{s.t.} \quad & 5x_1 + x_4 + x_5 = 25 \\ & 2x_2 + 2x_4 + x_5 = 15 \\ & 2x_3 + x_5 = 10 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

Its optimal value is 15.0 and its optimal solution is:

$$x = \begin{bmatrix} 2.5 \\ 0 \\ 0 \\ 2.5 \\ 10 \end{bmatrix}.$$

We see that basic variables are x_1, x_4 and x_5 . Hence:

$$B = \begin{bmatrix} 5 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 0.2 & -0.1 & -0.1 \\ 0 & 0.5 & -0.5 \\ 0 & 0 & 1 \end{bmatrix}$$

From there we get:

$$\begin{aligned} \hat{y}^\top &= c_B^\top B^{-1} = [1 \quad 1 \quad 1] \begin{bmatrix} 0.2 & -0.1 & -0.1 \\ 0 & 0.5 & -0.5 \\ 0 & 0 & 1 \end{bmatrix} = \\ &= [0.2 \quad 0.4 \quad 0.4] = \\ &= [1/5 \quad 2/5 \quad 2/5] \end{aligned}$$

Using this dual optimal solution we get following pricing/knapsack problem:

$$\begin{aligned} \max \quad & 0.2a_1 + 0.4a_2 + 0.4a_3 \\ \text{s.t.} \quad & 20a_1 + 35a_2 + 45a_3 \leq 100 \\ & a_1, a_2, a_3 \geq 0, \text{ integer} \end{aligned}$$

By solving this problem, we get new pattern:

$$A_5 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

with optimal value $\hat{z} = 1$. Since $1 - \hat{z} = 0 \geq 0$, we terminate the algorithm.

- (f) Write down the final optimal solution, the optimal basis, and the optimal objective value.

Solution:

Optimal value: 15.0

Optimal solution: $x_1 = 2.5, x_2 = 0, x_3 = 0, x_4 = 2.5, x_5 = 10$.

Basic Matrix: $B = \begin{bmatrix} 5 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$